WORKED SOLUTION

STEP III 2007, Question 3

Pure • Fibonacci Identities

STEP III 2007, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q3

The key idea

Every identity here is proved the same way: take the term with the largest index and split it using the recurrence, $F_m = F_{m-1} + F_{m-2}$. Doing that once turns the expression in (ii) into one with all indices shifted down by one; doing it twice brings you back to the same shape with the indices shifted down by two, which is exactly what an induction needs. For the divisibility in (iv), notice that $F_{2n-1} + F_{2n} = F_{2n+1}$, so F_{2n-1}^2 and F_{2n}^2 leave the same remainder on division by F_{2n+1}.

Solution

Part (i)

Adding consecutive terms from $F_1 = F_2 = 1$:

$$F_3 = 2, \quad F_4 = 3, \quad F_5 = 5, \quad F_6 = 8, \quad F_7 = 13, \quad F_8 = 21.$$

Part (ii)

Let $k \geq 1$, so that every term is defined. Replace the largest-index term F_{2k+3} by $F_{2k+2} + F_{2k+1}$:

$$\begin{aligned} F_{2k+3}F_{2k+1} - F_{2k+2}^2 &= (F_{2k+2} + F_{2k+1})F_{2k+1} - F_{2k+2}^2 \\ &= F_{2k+1}^2 + F_{2k+2}(F_{2k+1} - F_{2k+2}) \\ &= F_{2k+1}^2 - F_{2k+2}F_{2k}, \end{aligned}$$

since $F_{2k+2} - F_{2k+1} = F_{2k}$ by the recurrence. This is $-F_{2k+2}F_{2k} + F_{2k+1}^2$, as required.

Part (iii)

Write $P(n)$ for the statement $F_{2n+1}F_{2n-1} - F_{2n}^2 = 1$.

Base case. For $n = 1$: $F_3F_1 - F_2^2 = 2 \cdot 1 - 1^2 = 1$, so $P(1)$ holds.

Inductive step. Suppose $P(k)$ holds for some $k \geq 1$. By part (ii),

$$F_{2k+3}F_{2k+1} - F_{2k+2}^2 = F_{2k+1}^2 - F_{2k+2}F_{2k}.$$

> **Ask yourself** Part (ii) only shifted the indices down by one. How do we get all the way back to $P(k)$?

Apply the same trick once more, now splitting the largest-index term $F_{2k+2} = F_{2k+1} + F_{2k}$:

$$\begin{aligned} F_{2k+1}^2 - F_{2k+2}F_{2k} &= F_{2k+1}^2 - (F_{2k+1} + F_{2k})F_{2k} \\ &= F_{2k+1}(F_{2k+1} - F_{2k}) - F_{2k}^2 \\ &= F_{2k+1}F_{2k-1} - F_{2k}^2, \end{aligned}$$

using $F_{2k+1} - F_{2k} = F_{2k-1}$ (valid since $2k+1 \geq 3$). By the hypothesis $P(k)$ this equals 1. Chaining the two displays,

$$F_{2k+3}F_{2k+1} - F_{2k+2}^2 = 1,$$

which is $P(k+1)$. By induction, $P(n)$ holds for all $n \geq 1$.

Deduction. Rearranging $P(n)$,

$$F_{2n}^2 + 1 = F_{2n+1}F_{2n-1},$$

and F_{2n-1} is an integer (every F_m is, by the recurrence), so $F_{2n}^2 + 1$ is divisible by F_{2n+1}.

Part (iv)

By the recurrence, $F_{2n-1} = F_{2n+1} - F_{2n}$. Squaring,

$$F_{2n-1}^2 = F_{2n+1}^2 - 2F_{2n+1}F_{2n} + F_{2n}^2,$$

so

$$F_{2n-1}^2 + 1 = F_{2n+1}(F_{2n+1} - 2F_{2n}) + (F_{2n}^2 + 1).$$

The first bracket is a multiple of F_{2n+1} (its cofactor is an integer), and $F_{2n}^2 + 1$ is a multiple of F_{2n+1} by part (iii). Hence $F_{2n-1}^2 + 1$ is divisible by F_{2n+1}.

In fact we can find the quotient. Using $F_{2n}^2 + 1 = F_{2n+1}F_{2n-1}$,

$$F_{2n-1}^2 + 1 = F_{2n+1}(F_{2n+1} - 2F_{2n} + F_{2n-1}) = F_{2n+1}(2F_{2n-1} - F_{2n}),$$

and for $n \geq 2$ the bracket is $F_{2n-1} - F_{2n-2} = F_{2n-3}$. So $F_{2n-1}^2 + 1 = F_{2n+1}F_{2n-3}$, a close cousin of the identity in (iii).

Checks

With $n = 3$: $F_6^2 + 1 = 65 = 13 \times 5 = F_7F_5$, and $F_5^2 + 1 = 26 = 13 \times 2 = F_7F_3$. With $n = 4$: $F_7^2 + 1 = 170 = 34 \times 5 = F_9F_5$. A computer confirmed the identities in (ii) and (iii), both divisibility statements and the quotient F_{2n-3} for every n up to 37.

Common slips

- Trying the induction using (ii) alone: it shifts the indices by only one, and the resulting expression $F_{2k+1}^2 - F_{2k+2}F_{2k}$ is not yet the hypothesis. One more use of the recurrence is needed.
- Forgetting the base case, or checking it with the wrong indices ($n = 1$ involves F_1, F_2, F_3).

Beyond STEP

Part (iii) is half of Cassini's identity, $F_{m+1}F_{m-1} - F_m^2 = (-1)^m$, which drops out instantly from determinants: the matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^m$ has entries F_{m+1}, F_m, F_m, F_{m-1}, and its determinant is $(-1)^m$. Divisibility facts such as (iv) are the start of the arithmetic of linear recurrences, where congruences do most of the work. Davenport's *The Higher Arithmetic* is a good next step for the number theory, and Strang's *Introduction to Linear Algebra* for the matrix viewpoint.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.